Process of Decision Making, System, Modeling and Support in DSS

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Typical Business Decision Aspects

- Decision may be made by a group
- Group member biases
- Groupthink
- Several, possibly contradictory objectives
- Many alternatives
- Results can occur in the future
- Attitudes towards risk
- Need information
- Gathering information takes time and expense
- Too much information
- "What-if" scenarios
- Trial-and-error experimentation with the real system may result in a loss
- Experimentation with the real system - only once
- Changes in the environment can occur continuously
- Time pressure

- **How are decisions made???**
- **What methodologies can be applied?**
- **What is the role of information systems in supporting decision making?**

DSS

- ***Decision***
- ***Support***
- ***Systems***

Decision Making

- ***Decision Making*: a process of choosing among alternative courses of action for the purpose of attaining a goal or goals**

- ***Managerial Decision Making* *is synonymous with the whole process of management*** **(Simon, 1977)**

Decision Making versus Problem Solving

Simon's 4 Phases of Decision Making

1. **Intelligence**
2. **Design**
3. **Choice**
4. **Implementation**

***Decision making* and *problem solving* are interchangeable**

Systems

- **A *SYSTEM* is a collection of objects such as people, resources, concepts, and procedures intended to perform an identifiable function or to serve a goal**

- **System Levels (Hierarchy): All systems are *subsystems* interconnected through *interfaces***

The Structure of a System

Three Distinct Parts of Systems (Figure 2.1)

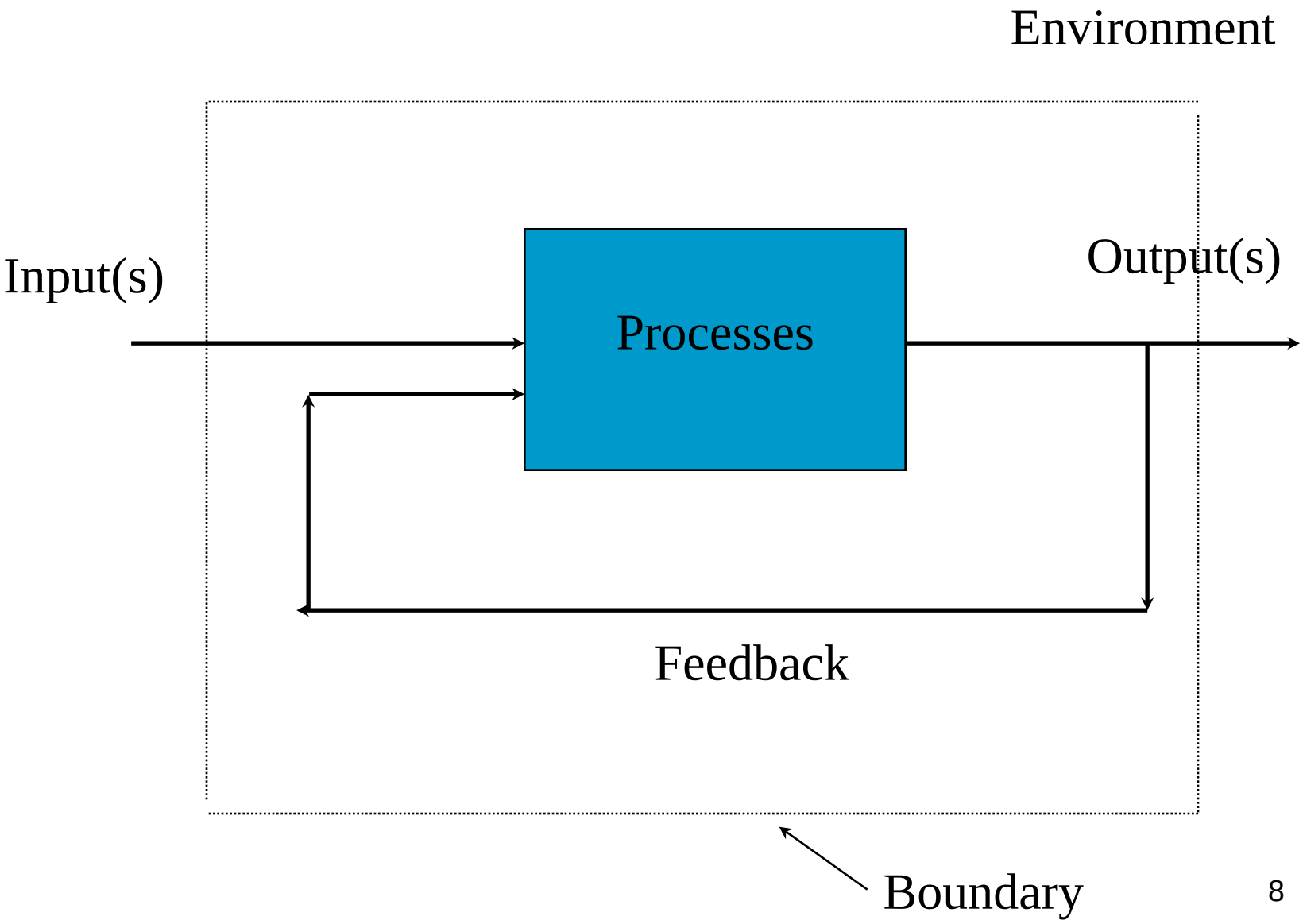
- Inputs
- Processes
- Outputs

Systems

- Surrounded by an environment
- Frequently include feedback

The decision maker is usually considered part of the system

System



- **Inputs are elements that enter the system**
- **Processes convert or transform inputs into outputs**
- **Outputs describe finished products or consequences of being in the system**
- **Feedback is the flow of information from the output to the decision maker, who may modify the inputs or the processes (closed loop)**
- **The Environment contains the elements that lie outside but impact the system's performance**

How to Identify the Environment?

Two Questions (Churchman, 1975)

1. Does the element matter relative to the system's goals? [YES]

2. Is it possible for the decision maker to significantly manipulate this element? [NO]

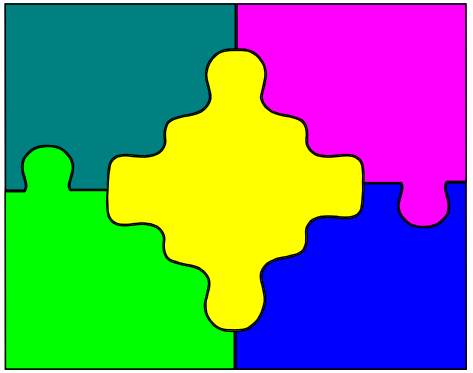
Environmental Elements Can Be

- Social
- Political
- Legal
- Physical
- Economical
- Often Other Systems

The Boundary Separates a System From Its Environment

Boundaries may be physical or nonphysical (by definition of scope or time frame)

Information system boundaries are usually by definition!



Closed and Open Systems

Defining manageable boundaries is *closing* the system

- **A Closed System is totally independent of other systems and subsystems**
- **An Open System is very dependent on its environment**

An Information System

- **Collects, processes, stores, analyzes, and disseminates information for a specific purpose**
- **Is often at the heart of many organizations**
- **Accepts inputs and processes data to provide information to decision makers and helps decision makers communicate their results**

System Effectiveness and Efficiency

Two Major Classes of Performance Measurement

- ***Effectiveness*** **is the degree to which goals are achieved**
  Doing the right thing!

- ***Efficiency*** **is a measure of the use of inputs (or resources) to achieve outputs**
  Doing the thing right!

- **MSS emphasize *effectiveness***
  Often: several non-quantifiable, conflicting goals

Models

- **Major component of DSS**
- **Use models instead of experimenting on the real system**

- **A *model* is a simplified representation or abstraction of reality.**
- **Reality is generally too complex to copy exactly**
- **Much of the complexity is actually *irrelevant* in problem solving**

Degrees of Model Abstraction

(Least to Most)



- ***Iconic (Scale) Model*: Physical replica of a system**
- ***Analog Model* behaves like the real system *but* does *not* look like it (symbolic representation)**
- ***Mathematical (Quantitative) Models* use mathematical relationships to represent complexity Used in most DSS analyses**

Benefits of Models

1. Time compression
2. Easy model manipulation
3. Low cost of construction
4. Low cost of execution (especially that of errors)
5. Can model risk and uncertainty
6. Can model large and extremely complex systems with possibly infinite solutions
7. Enhance and reinforce learning, and enhance training.

Computer graphics advances: more iconic and analog models (visual simulation)

The Modeling Process--
A Preview

- How Much to Order for the Ma-Pa Grocery?
- Bob and Jan: How much bread to stock each day?

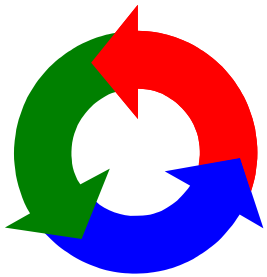
Solution Approaches

- Trial-and-Error
- Simulation
- Optimization
- Heuristics

The Decision-Making Process

Systematic Decision-Making Process (Simon, 1977)

- **Intelligence**
- **Design**
- **Choice**
- **Implementation**



(Figure 2.2)

Modeling is <u>Essential</u> to the Process

- **Intelligence phase**
  - **Reality is examined**
  - **The problem is identified and defined**
- **Design phase**
  - **Representative model is constructed**
  - **The model is validated and evaluation criteria are set**
- **Choice phase**
  - **Includes a proposed solution to the *model***
  - **If reasonable, move on to the**
- **Implementation phase**
  - **Solution to the original problem**

Failure: Return to the modeling process

Often Backtrack / Cycle Throughout the Process

The Intelligence Phase

Scan the environment to identify problem situations or opportunities

Find the Problem

- **Identify organizational goals and objectives**
- **Determine whether they are being met**
- **Explicitly define the problem**

Problem Classification

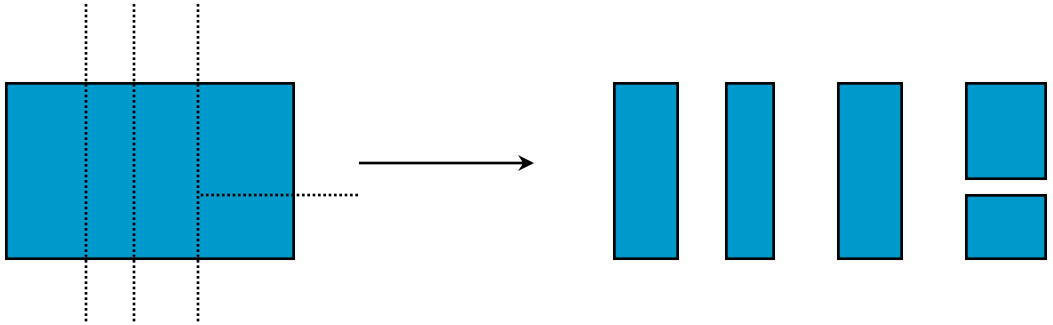
Structured versus Unstructured

Programmed versus Nonprogrammed Problems
Simon (1977)

Nonprogrammed Problems ⟷ **Programmed Problems**

- **<u>Problem Decomposition:</u> Divide a complex problem into (easier to solve) subproblems**

- **Some seemingly poorly structured problems may have some highly structured subproblems**



- **Problem Ownership**

  Outcome: <u>Problem Statement</u>

The Design Phase

- Generating, developing, and analyzing possible courses of action

Includes

- Understanding the problem
- Testing solutions for feasibility
- A model is constructed, tested, and validated

Modeling

- Conceptualization of the problem
- Abstraction to quantitative and/or qualitative forms

Mathematical Model

- **Identify variables**
- **Establish equations describing their relationships**
- **Simplifications through *assumptions***
- **Balance model simplification and the accurate representation of reality**

***Modeling*: an art and science**

Quantitative Modeling Topics

- **Model Components**
- **Model Structure**
- **Selection of a Principle of Choice (Criteria for Evaluation)**
- **Developing (Generating) Alternatives**
- **Predicting Outcomes**
- **Measuring Outcomes**
- **Scenarios**

Components of Quantitative Models

- **Decision Variables**
- **Uncontrollable Variables (and/or Parameters)**
- **Result (Outcome) Variables**
- **Mathematical Relationships**

  or

- **Symbolic or Qualitative Relationships**

Results of Decisions are Determined by the

- **Decision**
- **Uncontrollable Factors**
- **Relationships among Variables**

Result Variables

- **Reflect the level of effectiveness of the system**
- ***Dependent variables***

Decision Variables

- **Describe alternative courses of action**
- **The decision maker *controls* them**

Uncontrollable Variables or Parameters

- **Factors that affect the result variables**
- ***Not under the control* of the decision maker**
- **Generally part of the environment**
- **Some constrain the decision maker and are called *constraints***

Intermediate Result Variables

- **Reflect intermediate outcomes**

The Structure of Quantitative Models

- **Mathematical expressions (e.g., equations or inequalities) connect the components**
- **Simple financial model**
  $P = R - C$
- **Present-value model**
  $P = F / (1+i)^n$

LP Example

The Product-Mix Linear Programming Model

- **MBI Corporation**
- **Decision: How many computers to build next month?**
- **Two types of computers**
- **Labor limit**
- **Materials limit**
- **Marketing lower limits**

Constraint	CC7	CC8	Rel	Limit
Labor (days)	300	500	<=	200,000 / mo
Materials $	10,000	15,000	<=	8,000,000/mo
Units	1		>=	100
Units		1	>=	200
Profit $	8,000	12,000	Max	

Objective: Maximize Total Profit / Month

Linear Programming Model

- **Components**
  - ***Decision variables***
  - ***Result variable***
  - ***Uncontrollable variables (constraints)***

- ***Solution***
  - $X_1 = 333.33$
  - $X_2 = 200$
  - **Profit = $5,066,667**

Optimization Problems

- Linear programming
- Goal programming
- Network programming
- Integer programming
- Transportation problem
- Assignment problem
- Nonlinear programming
- Dynamic programming
- Stochastic programming
- Investment models
- Simple inventory models
- Replacement models (capital budgeting)

The Principle of Choice

- What criteria to use?
- Best solution?
- Good enough solution?

Selection of a Principle of Choice

Not the choice phase

A decision regarding the acceptability of a solution approach

- **Normative**
- **Descriptive**

Normative Models

- **The chosen alternative is demonstrably the best of all (normally a good idea)**
- ***Optimization* process**
- **Normative decision theory based on *rational* decision makers**

Rationality Assumptions

- **Humans are economic beings whose objective is to maximize the attainment of goals; that is, the decision maker is rational**
- **In a given decision situation, all viable alternative courses of action and their consequences, or at least the probability and the values of the consequences, are known**
- **Decision makers have an order or preference that enables them to rank the desirability of all consequences of the analysis**

Suboptimization

- **Narrow the boundaries of a system**
- **Consider a part of a complete system**
- **Leads to (possibly very good, but) non-optimal solutions**
- **Viable method**

Descriptive Models

- **Describe things as they are, or as they are believed to be**
- **Extremely useful in DSS for evaluating the consequences of decisions and scenarios**
- **No guarantee a solution is optimal**
- **Often a solution will be good enough**
- ***Simulation*: Descriptive modeling technique**

Descriptive Models

- Information flow
- Scenario analysis
- Financial planning
- Complex inventory decisions
- Markov analysis (predictions)
- Environmental impact analysis
- Simulation
- Waiting line (queue) management

Satisficing (Good Enough)

- **Most human decision makers will settle for a good enough solution**
- **Tradeoff: time and cost of searching for an optimum versus the value of obtaining one**
- **Good enough or *satisficing* solution may meet a certain goal level is attained**

(Simon, 1977)

Why Satisfice?
Bounded Rationality (Simon)

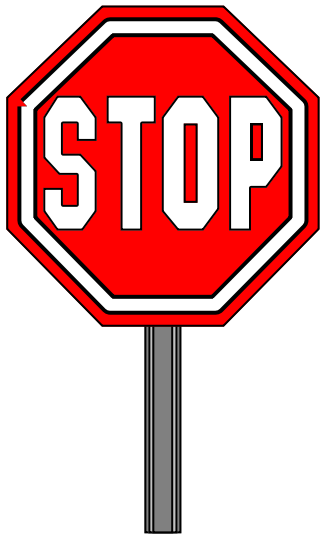
- Humans have a **limited capacity** for rational thinking
- Generally construct and analyze a **simplified model**
- **Behavior** to the simplified model may be rational
- But, the rational solution to the simplified model may **NOT BE** rational in the real-world situation
- Rationality is **bounded** by
  - limitations on human processing capacities
  - individual differences
- Bounded rationality: why many models are **descriptive, not normative**

Developing (Generating) Alternatives

- **In Optimization Models: Automatically by the Model!**

  Not Always So!

- **Issue: When to Stop?**



Predicting the Outcome of Each Alternative

- Must predict the future outcome of each proposed alternative
- Consider what the decision maker knows (or believes) about the forecasted results

- Classify Each Situation as Under
  - Certainty
  - Risk
  - Uncertainty

Decision Making Under Certainty

- *Assumes* complete knowledge available (deterministic environment)
- Example: U.S. Treasury bill investment

- Typically for structured problems with short time horizons
- Sometimes DSS approach is needed for certainty situations

Decision Making Under Risk (Risk Analysis)

- Probabilistic or stochastic decision situation
- Must consider several possible outcomes for each alternative, each with a probability
- Long-run probabilities of the occurrences of the given outcomes are assumed known or estimated

- Assess the (*calculated*) degree of risk associated with each alternative

Decision Making Under Uncertainty

- Several outcomes possible for each course of action
- *BUT* the decision maker does not know, or cannot estimate the probability of occurrence

- More difficult - insufficient information
- Assessing the decision maker's (and/or the organizational) attitude toward risk
- Example: poker game with no cards face up (5 card stud or draw)

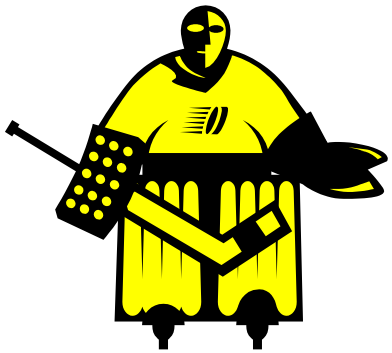
Measuring Outcomes

- **Goal attainment**
- **Maximize profit**
- **Minimize cost**
- **Customer satisfaction level (minimize number of complaints)**
- **Maximize quality or satisfaction ratings (surveys)**

Scenarios

Useful in

- **Simulation**
- **What-if analysis**



Importance of Scenarios in MSS

- **Help identify potential opportunities and/or problem areas**
- **Provide flexibility in planning**
- **Identify leading edges of changes that management should monitor**
- **Help *validate* major assumptions used in modeling**
- **Help check the sensitivity of proposed solutions to changes in scenarios**

Possible Scenarios

- **Worst possible (low demand, high cost)**
- **Best possible (high demand, high revenue, low cost)**
- **Most likely (median or average values)**
- **Many more**

- **The scenario sets the stage for the analysis**

The Choice Phase

- **The CRITICAL act - decision made here!**
- **Search, evaluation, and recommending an appropriate *solution* to the model**
- **Specific set of values for the decision variables in a selected alternative**

The problem is considered solved only after the recommended solution to the model is *successfully implemented*

Search Approaches

- **Analytical Techniques**
- **Algorithms (Optimization)**
- **Blind and Heuristic Search Techniques**

Evaluation: Multiple Goals, Sensitivity Analysis, What-If, and Goal Seeking

- Evaluation (with the search process) leads to a recommended solution
- Multiple goals
- Complex systems have multiple goals
  Some may conflict
- Typically, quantitative models have a single goal
- Can transform a multiple-goal problem into a single-goal problem

Common Methods

- **Utility theory**
- **Goal programming**
- **Expression of goals as constraints, using linear programming**
- **Point system**

- **Computerized models can support multiple goal decision making**

Sensitivity Analysis

- **Change inputs or parameters, look at model results**

Sensitivity analysis checks relationships

Types of Sensitivity Analyses

- **Automatic**
- **Trial and error**

Trial and Error

- Change input data and re-solve the problem
- Better and better solutions can be discovered
- How to do? Easy in spreadsheets (Excel)
  - What-if
  - Goal seeking

What-If Analysis

- **Figure 2.9 - Spreadsheet example of a what-if query for a cash flow problem**

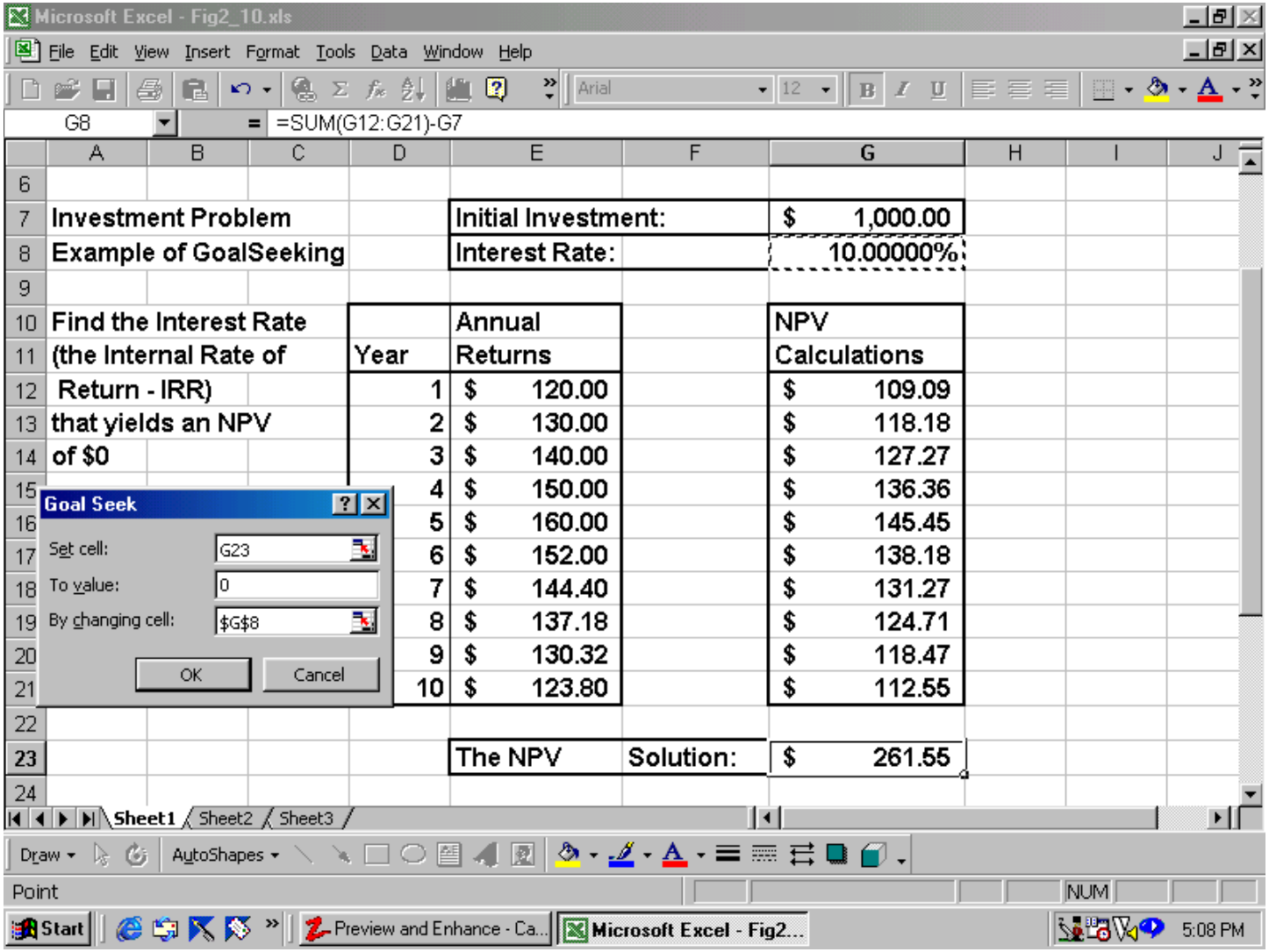
Goal Seeking

- **Backward solution approach**
- **Example: Figure 2.10**

  What interest rate causes an the net present value of an investment to break even?

- **In a DSS the what-if and the goal-seeking options *must* be easy to perform**

Goal Seeking



The Implementation Phase

There is nothing more difficult to carry out, nor more doubtful of success, nor more dangerous to handle, than to initiate a new order of things

(Machiavelli, 1500s)

***** The Introduction of a *Change* *****

Important Issues

- **Resistance to change**
- **Degree of top management support**
- **Users' roles and involvement in system development**
- **Users' training**

How Decisions Are Supported

Specific MSS technologies relationship to the decision making process (see Figure 2.10)

- **Intelligence: DSS, ES, ANN, MIS, Data Mining, OLAP, EIS, GSS**
- **Design and Choice: DSS, ES, GSS, Management Science, ANN**
- **Implementation: DSS, ES, GSS**

Alternative Decision Making Models

- Paterson decision-making process
- Kotter's process model
- Pound's flow chart of managerial behavior
- Kepner-Tregoe rational decision-making approach
- Hammond, Kenney, and Raiffa smart choice method
- Cougar's creative problem solving concept and model
- Pokras problem-solving methodology
- Bazerman's anatomy of a decision
- Harrison's interdisciplinary approaches
- Beach's naturalistic decision theories

Decision Styles

The manner in which decision makers

- Think and react to problems
- Perceive their
  - Cognitive response
  - Values and beliefs
- Varies from individual to individual and from situation to situation
- Decision making is a nonlinear process

The manner in which managers make decisions (and the way they interact with other people) describes their decision style

- There are dozens

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Some Decision Styles

- **Heuristic**
- **Analytic**
- **Autocratic**
- **Democratic**
- **Consultative (with individuals or groups)**
- **Combinations and variations**

- **For successful decision-making support, an MSS must fit the**
  - **Decision situation**
  - **Decision style**

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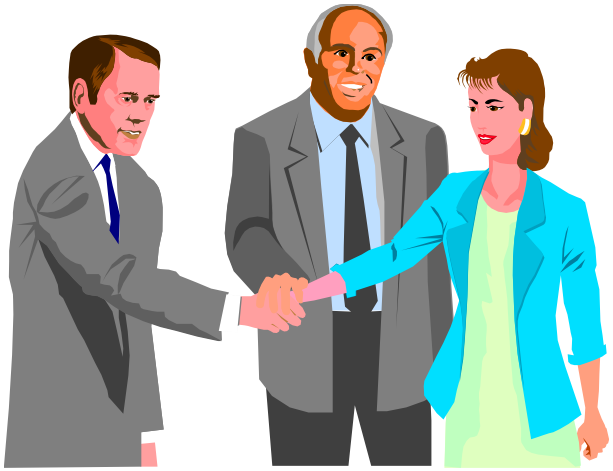
- **The system**
  - **should be flexible and adaptable to different users**
  - **have what-if and goal seeking**
  - **have graphics**
  - **have process flexibility**
- **An MSS should help decision makers use and develop their own styles, skills, and knowledge**
- **Different decision styles require different types of support**
- **Major factor: individual or group decision maker**

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The Decision Makers

- **Individuals**
- **Groups**





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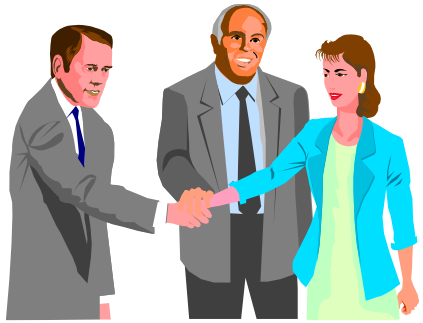
Individuals



- **May still have conflicting objectives**
- **Decisions may be fully automated**

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Groups



- Most major decisions made by groups
- Conflicting objectives are common
- Variable size
- People from different departments
- People from different organizations
- The group decision-making process can be very complicated
- Consider Group Support Systems (GSS)

- Organizational DSS can help in enterprise-wide decision-making situations

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Summary

- Managerial decision making is the whole process of management
- Problem solving also refers to opportunity's evaluation
- A system is a collection of objects such as people, resources, concepts, and procedures intended to perform an identifiable function or to serve a goal
- DSS deals primarily with open systems
- A model is a simplified representation or abstraction of reality
- Models enable fast and inexpensive experimentation with systems

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- **Modeling can employ optimization, heuristic, or simulation techniques**
- **Decision making involves four major phases: intelligence, design, choice, and implementation**
- **What-if and goal seeking are the two most common sensitivity analysis approaches**
- **Computers can support all phases of decision making by automating many required tasks**
- **Personality (temperament) influences decision making**
- **Gender impacts on decision making are inconclusive**
- **Human cognitive styles may influence human-machine interaction**
- **Human decision styles need to be recognized in designing MSS**

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